

Article

Preparation of PP-g-(AA-MAH) Fibers Using Suspension Grafting and Melt-Blown Spinning and its Adsorption for Aniline

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Received: 26 August 2020; Accepted: 18 September 2020; Published: 22 September 2020



Abstract: This paper uses polypropylene (PP) as the matrix and acrylic acid (AA) and maleic anhydride (MAH) as functional monomers to prepare PP-g-(AA-MAH) fibers by suspension grafting and melt-blown spinning technology that are easy to industrially scale-up. The fibers can be used to adsorb aniline. Results showed that the grafting ratio reached the maximum of 12.47%. The corresponding optimal conditions were grafting time of 3 h, AA: MAH = 0.75, total monomer content of 55%, benzoyl peroxide 1.4%, xylene concentration of 6 mL/g PP, and deionized water content of 8 mL/g PP. Owing to its good fluidity and thermal stability, the product of suspension grafting can be used for melt-blown spinning. Infrared spectroscopic and nuclear magnetic resonance spectroscopic analyses indicated that AA and MAH were successfully grafted onto PP fibers. After grafting, the hydrophilicity of PP-g-(AA-MAH) fiber increased. Therefore, it had higher absorptivity for aniline and the adsorption capacity could reach 42.2 mg/g at 45 min and pH = 7. Moreover, the PP-g-(AA-MAH) fibers showed good regeneration performance.

Keywords: polypropylene; suspension grafting; melt-blown spinning; adsorption; aniline

1. Introduction

Water discharged from industrial activities, if left untreated or not treated to the proper standard, inevitably causes water pollution, which is one of the major environmental issues of the day [1,2]. Aniline is one of the common organic compounds used as an intermediate in chemical industry, and hence its market demand has risen sharply [3–5]. This provides more channels for aniline to enter water bodies, thus causing environmental pollution. Since aniline does not readily undergo degradation, it accumulates in the environment. Moreover, being a carcinogenic, teratogenic, mutagenic, and highly toxic substance, aniline has a severely high risk of causing potential damage to the environment and organisms [6–8]. Therefore, aniline has been included in the list of preferentially controlled pollutants by Minister of Ecology and Environment of China, the United States Environmental Protection Agency, European Chemicals Agency, and other agencies [9–11].

At present, treatment methods for aniline-containing wastewaters mainly include advanced oxidation [12,13], membrane separation [14,15], extraction [16,17], biological methods [18], and adsorption methods [19–22]. Among them, the adsorption method that uses the interactions between adsorbent and adsorbate to remove pollutants from wastewater has attracted the attention of researchers, because of its simple operation, fast processing, and resourceful recovery [23]. Especially for treating sudden incidents of water pollution, adsorption method is considered as the first choice, owing to its high efficiency. Therefore, the development of adsorbents using a wide range of raw

materials, at low price, and achieving strong adsorption capacity through simple post-adsorption treatment of aniline wastewater has become the focus of research [24].

Polypropylene (PP) is acid- and alkali-resistant, very common material, cheap, and easy to process. After PP is spun by melt-blowing into fibers, it forms a three-dimensional network structure, with smaller mono-filament diameter, higher porosity, and larger specific surface area [25–27]. Therefore, it has been widely used as a matrix fiber. However, since PP is non-polar in nature, it does not possess polar and reactive groups along its molecular chains, which limits its ability to adsorb pollutants from wastewaters [28]. Our research group has been actively involved in the preparation of functionalized PP fibers with special properties through grafting, chemical modification, and other methods [29–31].

Suspension grafting [32] is a new, environment-friendly, and easy-to-industrialize method for grafting modification. This method usually employs water as the dispersion medium and polymer particles as the dispersed phase. It forms a suspension system, under the action of shear force provided by mechanical stirring. Each suspended polymer particle can be regarded as an independent “reaction bed”. Within this “reaction bed”, the functional monomers undergo graft polymerization under the action of initiators. This polymerization method has advantages of high reaction efficiency, few side reactions, and simple post-processing of products. Li et al. [33] prepared PP-g-PMMA with water as the dispersant and dibenzoyl peroxide as the initiator using the suspension grafting method. The addition of a second monomer, styrene, increased the grafting ratio of monomer to 24.5%. This indicated an effective improvement in the compatibility of PP/ASA mixture.

In this study, the amorphous region of PP was first expanded by quenching treatment, and then functional monomers like acrylic acid (AA) and maleic anhydride (MAH) were grafted onto the surface of PP by suspension grafting. Finally, the PP-g-(AA-MAH) fibers were prepared by melt-blown spinning technology. The adsorption and regeneration performance of PP-g-(AA-MAH) fibers for aniline in aqueous solution were investigated.

2. Experimental

2.1. Experimental Materials

PP resin especially for the melt-blown process was provided by Shanghai Expert New Material Co. Ltd. (Shanghai, China). Other chemical reagents, such as AA, MAH, xylene, benzoyl peroxide (BPO), acetone, absolute ethanol, sodium hydroxide, hydrochloric acid, etc. were all analytically pure and commercially available.

2.2. Preparation of PP-g-(AA-MAH) Fibers

A certain amount of PP resin and xylene were charged into a magnetically stirred reactor and the temperature was raised to 140 °C under stirring. Then, a certain amount of deionized water was rapidly pumped at normal temperature into the reactor to cool down the temperature of the reaction mixture. After this, the temperature of the reactor was maintained at 90 °C and then the mixture was allowed to swell for 30 min. Then, the monomers AA and MAH and BPO initiator were added to the reaction kettle through the high-pressure feed port and graft polymerization was carried out for a certain period of time under strong stirring. The obtained products were washed with deionized water and acetone to remove unreacted monomers and homopolymers. The washed product was dried and then passed through a melt-blown spinning machine [34] to obtain PP-g-(AA-MAH) fibers.

2.3. Calculation of Grafting Ratio

PP-g-(AA-MAH) fibers (0.5 g) were placed in a 250 mL round-bottom flask. Xylene (100 mL) and sodium hydroxide-ethanol solution (0.05 mol/L, 30 mL) were added in succession. The mixture was allowed to boil at 120 °C for 30 min. After the solution was cooled to room temperature, 3–5 drops of thymol blue indicator was added and then the mixture was titrated with hydrochloric acid-isopropanol

solution (0.05 mol/L) until the blue color just disappeared. The ungrafted PP fibers were used as the blank sample for comparison. The grafting ratio was calculated as follows:

$$G_p\% = \frac{(V_0 - V_1) \times C \times (98.06 + 72.06)}{3 \times m} \times 100\% \quad (1)$$

where $G_p(\%)$ denotes the grafting ratio; $V_0(L)$ and $V_1(L)$ represent the volumes of HCl-isopropanol solution consumed to titrate PP fibers and PP-g-(AA-MAH) fibers, respectively; $C(\text{mol/L})$ denotes the concentration of HCl-isopropanol solution; $m(\text{g})$ denotes the mass of the fiber sample; 98.06 g/mol and 72.06 g/mol are the molar masses of MAH and AA, respectively.

2.4. Characterization Methods

An X-ray diffractometer (X'TRA, ThermoFisher Scientific, Waltham, MA, USA) was used to characterize the crystal structures of sample before and after quenching. The fluidity and thermal stability of the suspension grafting product were studied by melt indexer (RL-Z1B1, SRD, Shanghai, China) and thermal analyzer (Pyris 1 DSC, PerKinElmer, Waltham, MA, USA), respectively. The functional groups of the modified fibers were analyzed using a Fourier transform infrared spectrometer (Nexus 670, Nicolet, Waltham, MA, USA) and nuclear magnetic resonance spectrometer (DRX500, BRUKER, Karlsruhe, Germany).

2.5. Hydrophilicity of PP-g-(AA-MAH) Fiber

The Washburn equation [35] can determine the contact angle of a dynamic liquid by measuring the speed at which the liquid penetrates into the filled powder (Equation (2)). However, the penetration height of wetting liquid is difficult to determine. As the penetration height changes, the air in the capillary tube gets compressed and the air pressure increases [36]. Therefore, Equation (2) can be revised by the relationship between osmotic pressure and height and Equation (3) is obtained.

Let $K_\theta = \frac{\beta\gamma_{LV}\cos\theta}{\eta}$, K_θ can be obtained from the linear relationship between $(\Delta P)^2$ and t .

In this study, the water contact angle of degreased cotton with good hydrophilicity was set to 0, that is, when $\theta = 0$, then the equation becomes $K_0 = \frac{\beta\gamma_{LV}}{\eta}$. Since the bulk densities of all the test samples was almost the same and deionized water was used as the wetting liquid, the β , η , and γ_{LV} values of the test samples were approximately the same. Due to these similarities, Equation (4) can be obtained for calculating the water contact angle on the surface of tested fiber. The schematic diagram of the self-made device is shown in Figure 1.

$$h^2 = \frac{R_{eff} \gamma_{LV} \cos \theta}{2\eta} t \quad (2)$$

$$(\Delta P)^2 = \frac{\beta\gamma_{LV} \cos \theta}{\eta} t \quad (3)$$

$$\theta = \arccos\left(\frac{K_\theta \eta}{\beta\gamma_{LV}}\right) = \arccos\left(\frac{K_\theta}{K_0}\right) \quad (4)$$

where h denotes the penetration height of the wetting liquid in time t ; R_{eff} denotes the effective capillary radius; γ_{LV} denotes the surface tension of the wetting liquid; θ denotes the water contact angle; η denotes the viscosity of the wetting liquid; t denotes the penetration time; ΔP denotes the change in osmotic pressure, and β denotes the parameters related to the stacking mode of the stacked bed.

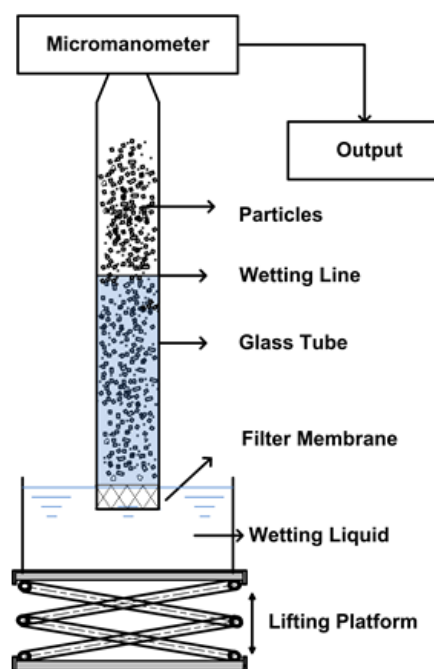


Figure 1. The schematic diagram of the self-made device.

2.6. Adsorption and Desorption of Aniline by PP-g-(AA-MAH) Fiber

Ten portions of potassium nitrate (0.01 mol/L, 20 mL) were taken and each portion was transferred to a conical flask. It was purged with N_2 to remove CO_2 . The pH was adjusted to values between 2–10 using 0.01 mol/L HCl and 0.01 mol/L NaOH solutions. The pH value was recorded as pH_0 . Then, PP-g-(AA-MAH) fibers (0.2 g) was added to the conical flask, flask was sealed, and shaken for 36 h in a water bath at room temperature. Thereafter, the fibers were taken out and the pH value of solution was measured, which was denoted as pH_f . The pH_0 values were plotted along the X axis and $pH_0 - pH_f$ values along the Y axis, to fit a straight line. The point of intersection of the straight line and the X axis was recorded as the point of zero charge of the PP-g-(AA-MAH) fiber.

Deionized water was used to prepare a mono-contaminated solution containing a certain concentration of aniline and the pH value was adjusted with 0.01 mol/L HCl and 0.01 mol/L NaOH solutions. After adsorption at 25 °C under shaking, the fibers were taken out. After allowing to stand for 5 min, the aniline concentration was determined, and the adsorption capacity was calculated from equation.

$$Q = \frac{(C_0 - C_1)V}{m} \quad (5)$$

where Q (mg/g) denotes the adsorption capacity; C_0 (mg/L) and C_1 (mg/L) denote the concentrations of aniline in the solution before and after adsorption; V (L) denotes the volume of the solution; and m (g) is the mass of fiber as adsorbent.

The adsorption kinetics were fitted with pseudo-first-order (Equation (6)) and pseudo-second-order (Equation (7)) models.

$$\ln(Q_e - Q_t) = \ln Q_e - K_1 t \quad (6)$$

$$\frac{t}{Q_t} = \frac{1}{K_2 Q_e^2} + \frac{t}{Q_e} \quad (7)$$

where Q_e (mg/g) and Q_t (mg/g) are the amount of adsorbed aniline at equilibrium and at an arbitrary time t (min), respectively; and K_1 (1/min) and K_2 (mg/g/min) are the pseudo-first-order and pseudo-second-order rate constants of adsorption, respectively.

The adsorption isotherm was analyzed with the Langmuir and Freundlich models:

$$\frac{C_e}{Q_e} = \frac{C_e}{Q_m} + \frac{1}{K_L Q_m} \quad (8)$$

$$\ln Q_e = \ln K_F + \frac{1}{n} \ln C_e \quad (9)$$

where C_e (mg/L) is the concentration of aniline in solution at equilibrium; Q_e (mg/g) is the equilibrium adsorption capacity; Q_m (mg/g) is the maximum adsorption capacity; K_L (L/mg) is the Langmuir constant; K_F (L/mg) is the Freundlich constant indicative of the adsorption capacity; n is a parameter related to the sorption intensity.

The adsorption-saturated PP-g-(AA-MAH) fiber was desorbed by shaking in 0.5 mol/L HCl solution for a certain period of time. The concentrations of aniline in the solution were measured at different times and the desorption ratios were calculated from Equation (10) [37].

$$D = \frac{CV}{mQ} \times 100\% \quad (10)$$

where D (%) denotes the desorption ratio; C denotes the concentration of aniline in the eluent (mg/L); Q is the adsorption capacity of the fiber (mg/g); V is the volume of the eluent (L), and m denotes mass of fiber (g).

3. Results and Discussion

3.1. The Mechanism of Suspension Grafting Polymerization of Two Monomers

The schematic diagram of suspension grafting polymerization of AA and MAH monomers [29] is shown in Figures 2 and 3. In the suspension grafting reaction system, under the action of shear force provided by high-speed stirring, each PP resin particle swelled up and was surrounded by xylene. It dispersed into an independent “reaction bed” in deionized water as the dispersant. The grafting polymerization in each “reaction bed” mainly occurred in the amorphous regions of PP resin. In this study, PP resin was quenched before the graft polymerization, in order to enlarge the amorphous region and PP transformed from a resin to a particle with surface micropores. The swelling effect could not only expand the amorphous region further but also enabled the BPO initiator and AA and MAH monomers to diffuse through the surface and into the pores of PP particles more uniformly. Especially, under the action of xylene, they could migrate to the internal micropores of PP more conveniently.

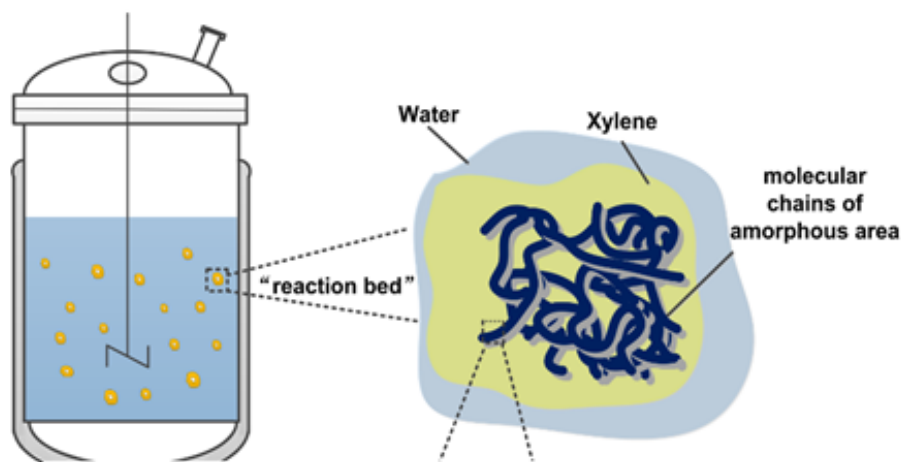


Figure 2. The suspension grafting reaction system.

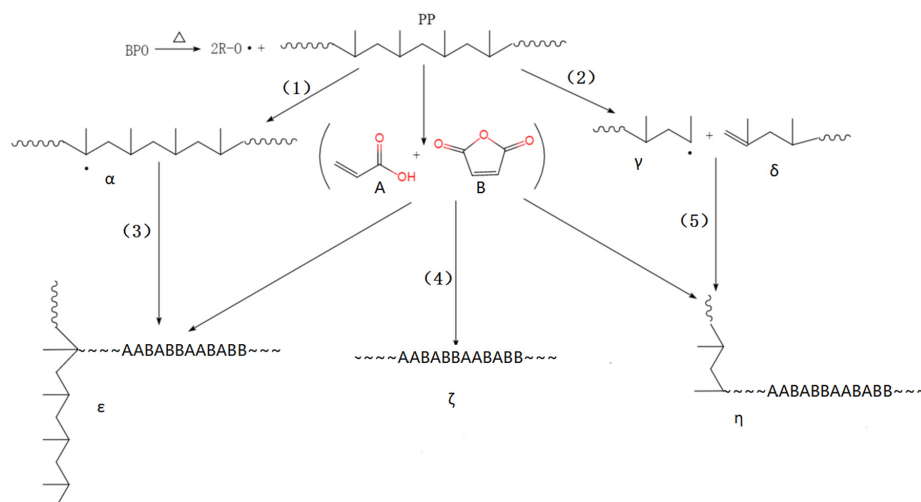


Figure 3. The mechanism of suspension grafting polymerization of AA and MAH monomers.

Generally, the initiator first decomposes to form primary free radicals. The primary free radicals can generate PP macromolecular free radicals denoted as α , by capturing α -H from the PP molecular chains (Reaction 1). The macromolecular free radicals, α , react with monomers to form the target product, ϵ (Reaction 3). However, side reactions like homo-polymerization of monomers (Reaction 4) and PP degradation (Reaction 2), also occur during this process. In the presence of AA, AA first attacks the PP macromolecular radicals α during the grafting polymerization, since AA is an electron provider and has higher reactivity. It can rapidly react with the PP macromolecular radicals α . Moreover, it inhibits side reactions like degradation to form stable PP-AA macromolecular free radicals. The macromolecular free radicals further react with AA or MAH, which is the main reaction pathway. At the same time, due to low reaction temperature, it becomes difficult for the degradation products to react with the monomer (Reaction 5).

3.2. Effect of Quenching Treatment

The PP used in this research is a special raw material for melt-blown process. It has high crystallinity, but the suspension grafting modification is mainly carried out in the amorphous regions of PP. Therefore, the PP melt can be quenched to reduce its crystallinity, which makes the grafting polymerization easier. It can be seen from the X-ray diffraction patterns of PP before and after the quenching treatment in Figure 4 that the dispersion peak intensity (shaded area in the figure) of the amorphous region of the quenched PP was significantly greater than that of the original PP. Since the quenching process greatly shortens the crystallization time of the PP melt, some of the PP macromolecular chains have no time to move, transform from the disordered coils and then grow into crystals, which results in lower crystallinity.

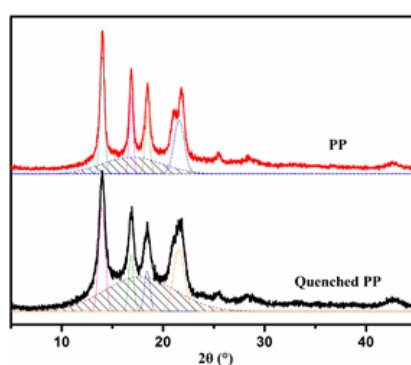


Figure 4. X-ray diffraction patterns of PP before and after the quenching treatment.

3.3. Effects of Different Parameters on Grafting Ratio in Suspension Grafting Polymerization

The main factors affecting grafting in suspension grafting polymerization include the grafting time, the AA/MAH ratio, the total amount of monomers, and the dosages of BPO, xylene, and deionized water. Therefore, effects of the six factors on the grafting ratio were all investigated and the results are shown in Figure 5.

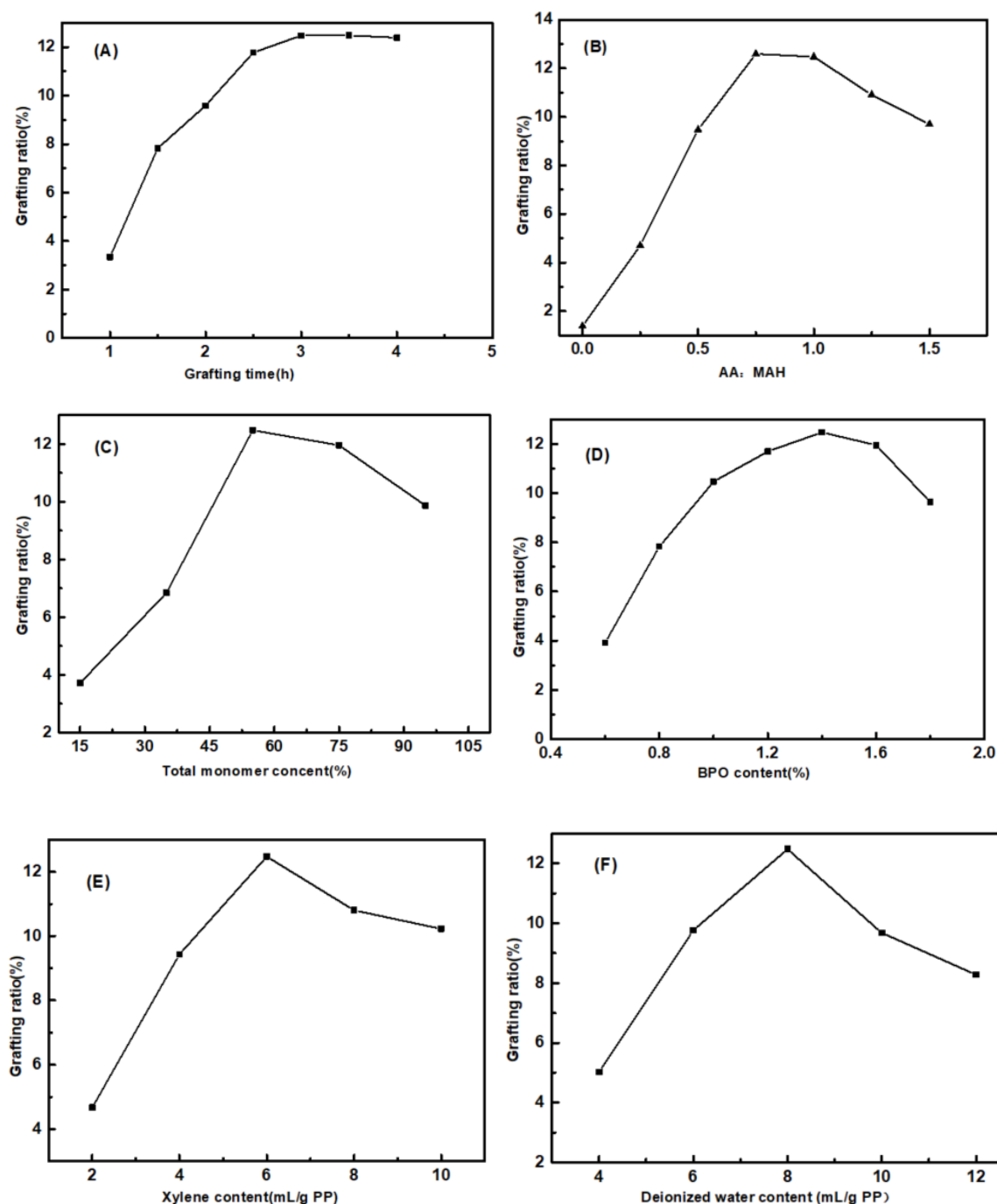


Figure 5. Effects of grafting time (A), AA/MAH ratio (B), total monomer content (C), and the dosages of BPO (D), xylene (E), and deionized water (F) on grafting ratio.

It can be seen from Figure 5A that the grafting ratio increased rapidly with increase in grafting time and it reached a stable state after three hours. The half-life period of the initiator BPO at 90 °C is 1 h [29]. In the initial stages of the reaction, BPO decomposed into a large number of primary free

radicals, which rapidly initiated the grafting reaction. In addition, the monomer concentration was also high at this time, due to which the grafting ratio increased rapidly. After 3 h, BPO was basically decomposed and the monomer concentration was also lower, exhibiting a more stable grafting ratio.

From Figure 5B, it is evident that when the mass fraction of AA was 0, the grafting ratio was only 1.38%. Subsequently, as the AA/MAH ratio increased, the grafting ratio increased at first and then decreased. With increase in the amount of monomer AA added, the concentration of macromolecular free radicals with AA chains increased. This increased the probability of free radicals contacting with PP. In addition, the monomer MAH could not undergo self-polymerization, due to its steric hindrance. When AA was added, it could form a copolymer with MAH and then graft with the PP segment [38], due to which the grafting ratio increased. However, as the amount of AA added continued to increase, the probability of capturing primary free radicals increased, thereby forming a greater number of AA-containing macromolecular free radicals. The probability of termination due to recombination between these free radicals increased, which resulted in lower grafting ratio. In addition, more the number of AA molecules, more was the probability of AA initiation, which then reduced the probability of formation of macromolecular free radicals of PP, thereby affecting the grafting ratio.

It can be seen from Figure 5C that as the total amount of monomers AA and MAH increased, the grafting ratio increased at first and then decreased. Higher monomer concentration increased the effective collisions between the monomers and PP macromolecular radicals, thereby increasing the grafting ratio. However, grafting polymerization mainly occurred in the amorphous regions of PP. Although the amorphous regions of PP expanded after quenching, the reaction space was still limited. Too high monomer concentration led to more homopolymerization as the side reaction. Additionally, the mass transfer of monomer and initiator were also affected, so the grafting ratio decreased.

With an increase in the amount of initiator BPO, the grafting ratio first increased and then decreased. When the amount of added BPO accounted for 1.4% of PP, the grafting ratio reached the maximum value of 12.47% (Figure 5D). On heating, BPO first decomposed into benzoyl radicals. Due to steric hindrance, benzoyl radicals were further decomposed into benzene radicals with stronger capability for H atom. Benzene radicals flowed and diffused slowly in the PP particles within limited space, depriving the α -H from the PP chains to form PP macromolecular radicals and initiating monomer polymerization. Therefore, by appropriately increasing the amount of BPO, a greater number of benzene radicals could be generated. This increased the number of monomers participating in the grafting reaction and increased the grafting ratio. However, excessively high concentration of benzene radicals could cause explosive polymerization and lead to an increase in homopolymerization side reactions. This would again affect the progress of the grafting reaction and also result in degradation of PP [39].

It can be seen from Figure 5E that, with the increase in amount of xylene, the grafting ratio increased rapidly and then decreased slightly. With xylene concentration of 6 mL/g PP, the grafting ratio reached the maximum value of 12.47%. The monomers used in this study possessed polar groups, and hence their solubility parameters were quite different from that of PP, which implied lower swelling capability of PP. The addition of certain amount of xylene helped to increase the swelling degree of PP and provided greater space for the grafting reaction. Meanwhile, the addition of xylene also kept the concentration of water-soluble monomers at the interface to a low level. This effectively reduced homo-polymerization as side reaction and increased the grafting ratio. However, too much xylene readily formed a film covering the surface of PP, which hindered the grafting reaction between monomer and PP and affected the grafting ratio.

Figure 5F shows that as the amount of deionized water as dispersant increased, the grafting ratio increased rapidly and then decreased. As the amount of deionized water reached 8 mL/g PP, the grafting ratio reached the maximum value. If the amount of deionized water added was too less, the dispersion of PP into multiple independent “reaction beds” became difficult, since PP particles agglomerated in the reactor to form agglomerates, resulting in a low grafting ratio. With too much of deionized water, although the PP particles could form numerous independent “reaction beds”,

the concentration of monomer dissolved in the water decreased. Hence, the monomer concentration gradient between the dispersion medium and the interfacial reaction layer became smaller. As a result, it was difficult for the monomer molecules to enter the interfacial layer and participate in the grafting polymerization reaction, due to which the grafting ratio decreased.

Based on the aforementioned study of the effect of parameters of suspension graft polymerization on the grafting rate, the optimal combination of parameters for suspension graft polymerization are presented in Table 1.

Table 1. The optimal combination of parameters for suspension graft polymerization.

PP	Monomers		Xylene	Deionized Water	BPO	Grafting Time	Grafting Ratio
	AA	MAH					
200 g	47 g	63 g	1.2 L	1.6 L	2.8 g	3 h	12.47%

3.4. Effect of Suspension Grafting on the Performance of Melt-Blown Spinning Product

Melt-blown spinning has strict requirements of fluidity and thermal stability of raw materials. PP resin itself has greater fluidity, but after its modification with polar groups, the interactions between molecular chains increase and the fluidity is affected. In addition, in the melt-blown spinning process, the raw materials are melted at high temperature and extruded through the screw. Hence, only better thermal stability can ensure that no decomposition of the raw materials occurs during the process. Therefore, the melt flow rate (MFR) and thermogravimetric (TG) of PP and PP-g-(AA-MAH) were investigated to determine whether it can be spun into fibers. The results are presented in Figure 6.

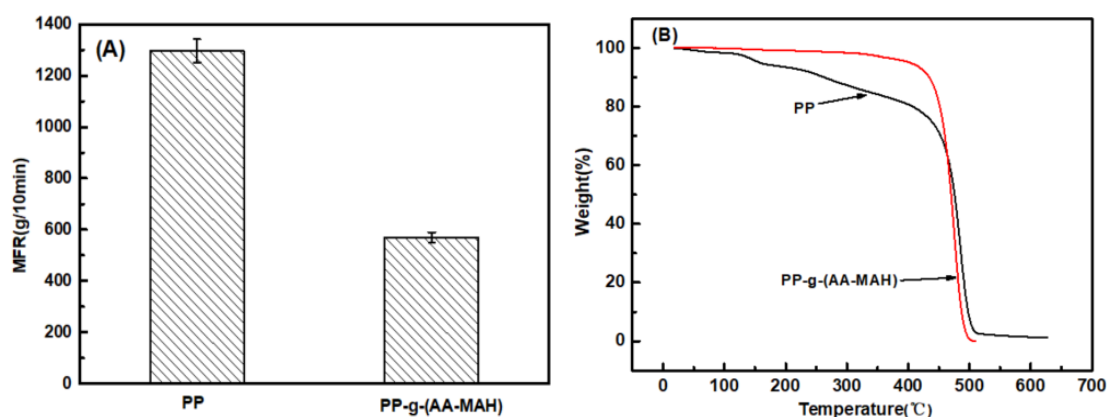


Figure 6. MFR (A) and TG (B) of PP and PP-g-(AA-MAH) fibers.

It can be seen from Figure 6A that MFR (570 g/10 min) of the product PP-g-(AA-MAH) of suspension grafting was lower than that of the PP raw material (1296 g/10 min), implying decrease in fluidity. However, the MFR was still higher than that required for the melt-blown process of PP melt (400 g/10 min). The decrease in MFR could be attributed mainly to the strengthening of hydrogen bonds between the grafted polar chains [29,34]. Compared with the TG curves of unmodified PP in Figure 6B, the product of suspension grafting PP-g-(AA-MAH) had better thermal stability. PP-g-(AA-MAH) showed no significant weight loss before 250 °C, and hence the melt-blown spinning at maximum temperature of 220 °C basically did not cause decomposition.

3.5. Characterization of Fibers

3.5.1. Fourier Transform Infrared Analysis (FT-IR)

PP and its products of suspension grafting - PP-g-AA, PP-g-MAH, and PP-g-(AA-MAH) were spun by melt-blowing and processed into fibers. Their FTIR spectra (the total monomer contents

were the same during the suspension grafting process) are shown in Figure 7. The fibers of modified PP-g-AA, PP-g-MAH, and PP-g-(AA-MAH) showed unique characteristic absorption peaks of PP, viz. symmetrical (2924 cm^{-1}) and asymmetrical (2842 cm^{-1}) C-H stretching vibrations of $-\text{CH}_3$, the flexural vibrations of $-\text{CH}_2-$ at 1380 cm^{-1} , and absorption of C-H at 1450 cm^{-1} . Compared with that of PP fiber, the FTIR spectrum of PP-g-AA fiber showed peak for C=O stretching vibrations of the carboxyl group (1720 cm^{-1}), indicating the successful grafting of AA onto the PP chain. Compared to the spectrum of PP fiber, PP-g-MAH showed a peak at 1780 cm^{-1} , which was attributed to the symmetric and asymmetric vibrations of C=O of acid anhydride. The peak at 1720 cm^{-1} was ascribed to the C=O of carboxylic acid, formed by the hydrolysis of anhydride. The FTIR spectra of PP-g-(AA-MAH), PP-g-AA, and PP-g-MAH fibers all showed peaks for stretching vibrations of C=O of the carboxyl group at 1720 cm^{-1} . The difference was that the FTIR spectrum of PP-g-(AA-MAH) had stronger intensity for C=O peak, indicating that its carboxyl content was higher than those of PP-g-AA and PP-g-MAH fibers. However, the FTIR spectrum of PP-g-(AA-MAH) fiber did not show peaks for symmetrical and asymmetrical vibrations of C=O of acid anhydride in the PP-g-MAH spectrum. This was due to the fact that MAH was grafted on to PP in the form of maleic acid [39,40]. FTIR results confirmed the successful grafting of AA and MAH onto PP fibers.

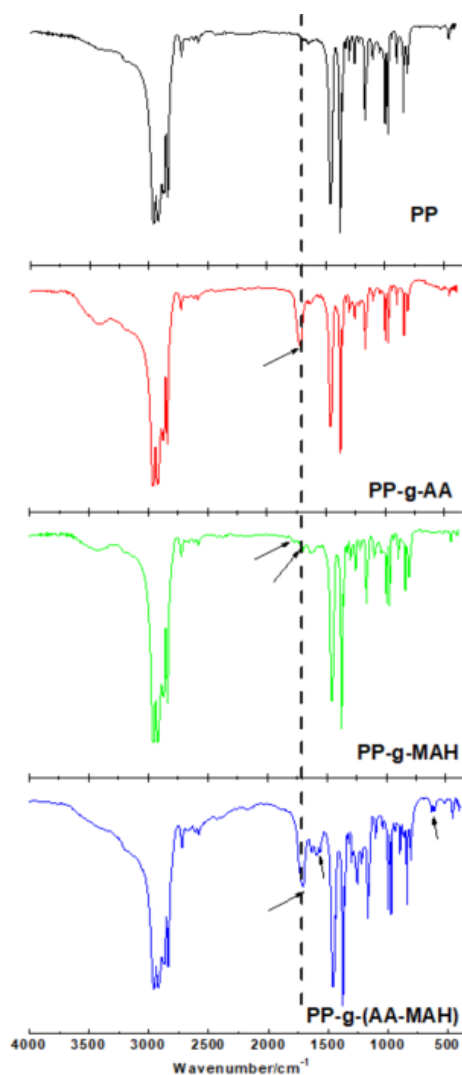


Figure 7. FTIR spectra of PP, PP-g-AA, PP-g-MAH and PP-g-(AA-MAH) fibers.

3.5.2. ^1H -NMR Analysis

The carboxyl group in PP-g-(AA-MAH) fiber was linked to $-\text{CH}-$. From the ^1H -NMR analysis (Figure 8), it was evident that compared with PP fiber, PP-g-(AA-MAH) fiber showed a chemical shift at around 2.3 ppm [41], indicating that the carboxyl groups were successfully grafted onto the surface of PP fibers.

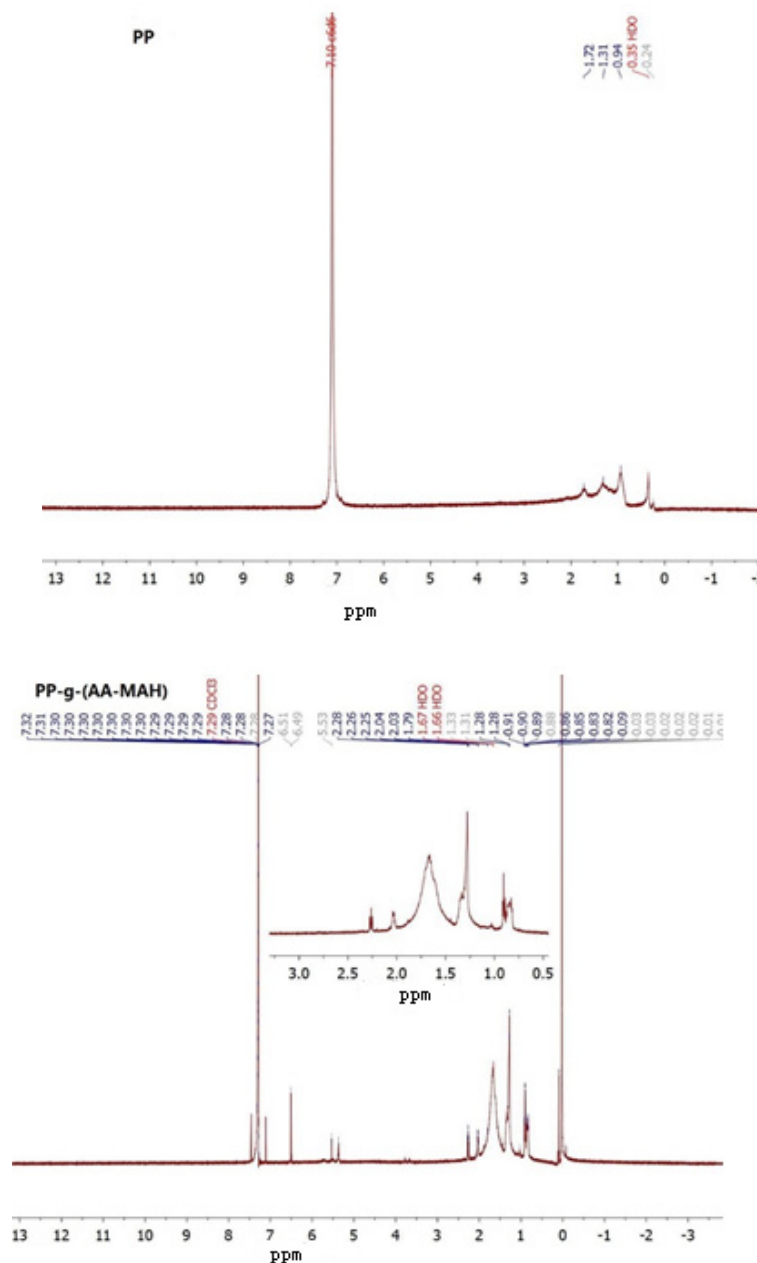


Figure 8. ^1H -NMR analysis of PP and PP-g-(AA-MAH) fibers.

3.5.3. Analysis on Hydrophilicity

Figure 9 showed a linear fitting relationship of $(\Delta P)^2$ of degreased cotton, PP-g-(AA-MAH) fiber and PP fiber with t . It could be seen from the figure that the K_0 of degreased cotton was 45.66, whereas K_0 of PP-g-(AA-MAH) fiber and PP fiber were 17.95 and 3.19, respectively. According to Equation (4), the calculated water contact angles of PP-g-(AA-MAH) fiber and PP fiber were 66.85° and 85.98° , respectively, which indicated increased hydrophilicity of PP-g-(AA-MAH) fibers.

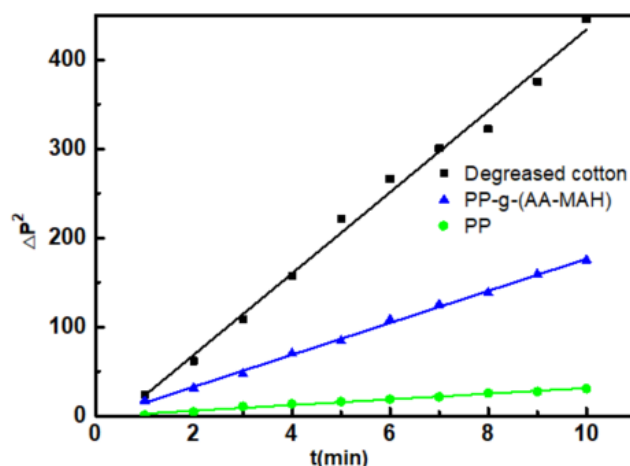


Figure 9. Linear fitting relationship of $(\Delta P)^2$ of degreased cotton, PP-g-(AA-MAH) and PP fiber with t .

3.6. Investigation of Aniline-Adsorption Performance of PP-g-(AA-MAH) Fiber before and after Regeneration

3.6.1. Effect of pH on the Adsorption Capacity

Solution pH value influences the surface properties of PP-g-(AA-MAH) fibers and the state of aniline in water, which in turn has an effect on the adsorption. Figure 10 shows the effect of pH value on the adsorption capacity of PP-g-(AA-MAH) fibers. The adsorption capacity first increased and then decreased in the pH range from 3 to 10. The adsorption capacity reached the maximum value of 48.3 mg/g, when the solution was neutral.

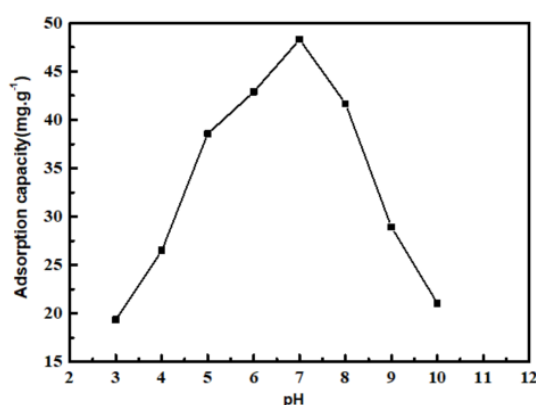


Figure 10. The effect of the pH value on the adsorption capacity.

Aniline is weakly alkaline and has pK_b value (dissociation constant) of 9.4. In other words, aniline exists as $C_6H_5NH_3^+$ below 9.4 pH and in molecular form above 9.4 pH [20,21]. Combining this with the point of zero charge (3.02) of the PP-g-(AA-MAH) fiber in the aqueous solution (from Figure 11), the surface of PP-g-(AA-MAH) fiber was protonated and positively charged below 3.02. Therefore, the electrostatic repulsion between aniline (in the form of $C_6H_5NH_3^+$) and positively charged surface led to its poor adsorption. Above 3.02, the surface of PP-g-(AA-MAH) fiber was negatively charged due to deprotonation. The number of $-COO^-$ groups on the surface increased and the electrostatic attraction was strengthened. Hence, the adsorption capacity of PP-g-(AA-MAH) fibers towards aniline gradually increased between pH 3~7. As pH value continued to rise and the aqueous solution turned alkaline, the reduction of $C_6H_5NH_3^+$ led to the reductions in adsorption capacity, although the number of $-COO^-$ groups increased. Above pH 9.4, aniline mainly existed in the molecular form and had lower solubility, which was not conducive for adsorption, and hence the adsorption capacity decreased further.

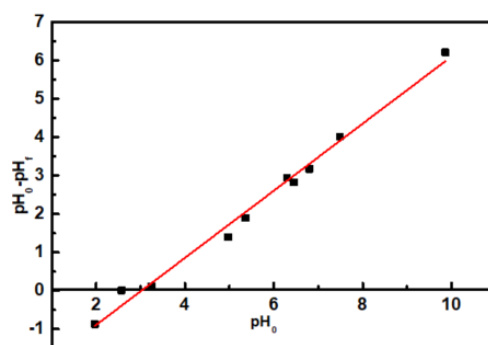


Figure 11. The point of zero charge of PP-g-(AA-MAH) fiber in the aqueous solution.

3.6.2. Adsorption Kinetics

Under the conditions of 25 °C, pH = 7, and initial aniline concentration of 300 mg/L, the adsorptivities of aniline on PP-g-(AA-MAH) fibers at different adsorption times were investigated, and the kinetics was studied, as shown in Figure 12. Figure 12A showed that PP-g-(AA-MAH) fibers had faster adsorptivity for aniline within the first 45 min, with adsorption capacity reaching up to 42.2 mg/g. Then, the adsorption capacity increased slowly until it reached saturation as 48.3 mg/g at 120 min. The faster rate of adsorption within the first 45 min was the result of the large number of active sites on the surface of PP-g-(AA-MAH) fibers. With the prolongation of adsorption time, the number of active sites on the surface of PP-g-(AA-MAH) fibers gradually decreased along with decrease in aniline concentration. As a result, the driving force for adsorption obviously weakened and the adsorption capacity increased slowly until it reached saturated adsorption. From the linear fitting results (Figure 12B,C), The R^2 of the pseudo-first-order kinetic model is 0.9734, while for the pseudo-second-order kinetic model, R^2 is 0.9955. Hence, the adsorption for aniline is better fitted by the pseudo-second-order kinetic model, indicating that the adsorption is controlled by chemical phenomena [42]. Hence, the electrostatic attraction is the main driving force for aniline adsorption on PP-g-(AA-MAH) fibers.

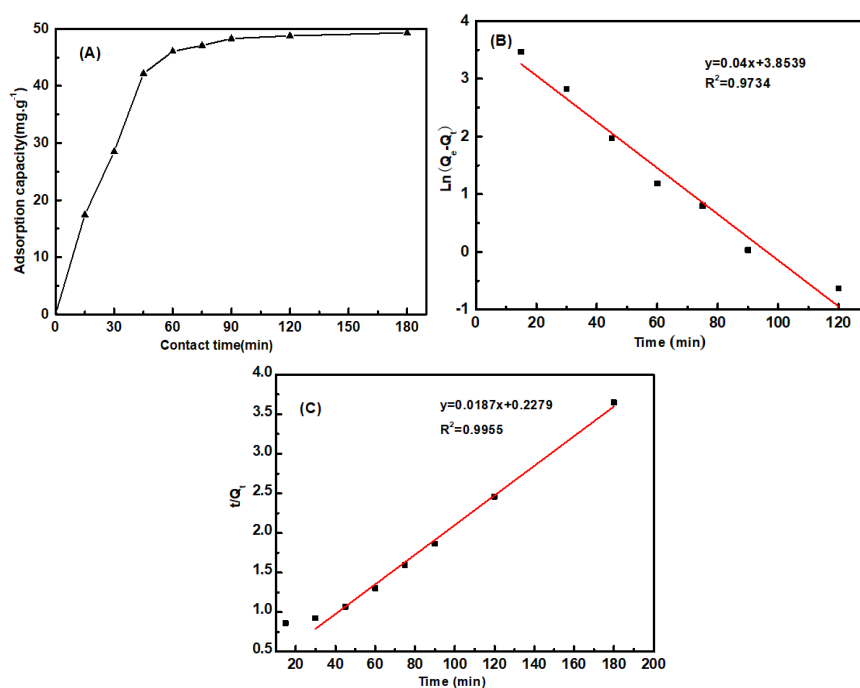


Figure 12. The effect of contact time on adsorption capacity (A) and the linearized pseudo-first-order (B) and pseudo-second-order (C) kinetics.

3.6.3. Adsorption Isotherm

The adsorption isotherm was drawn based on the results obtained using different initial aniline concentrations. As can be seen from Figure 13A, the adsorption capacity increased with the increase in initial aniline concentration, and eventually became stable. With the increase in aniline concentration, the probability of collision between functional groups on fibers and aniline increased, and so the adsorption capacity increased. However, the number of functional groups was limited, and the adsorption capacity eventually tended to be stable. The linearized versions of the two models are illustrated in Figure 13B,C and the values of the isotherm parameters are listed in Table 2. It was found that the R^2 value of Langmuir and Freundlich was 0.9925 and 0.9791, respectively, and the adsorption for aniline of PP-g-(AA-MAH) fibers is better fitted by the Langmuir model. The uptake of aniline occurs on the homogeneous surface of PP-g-(AA-MAH) fibers by monolayer adsorption. The maximum theoretical adsorption capacity for aniline of PP-g-(AA-MAH) fibers is 62.7 mg/g.

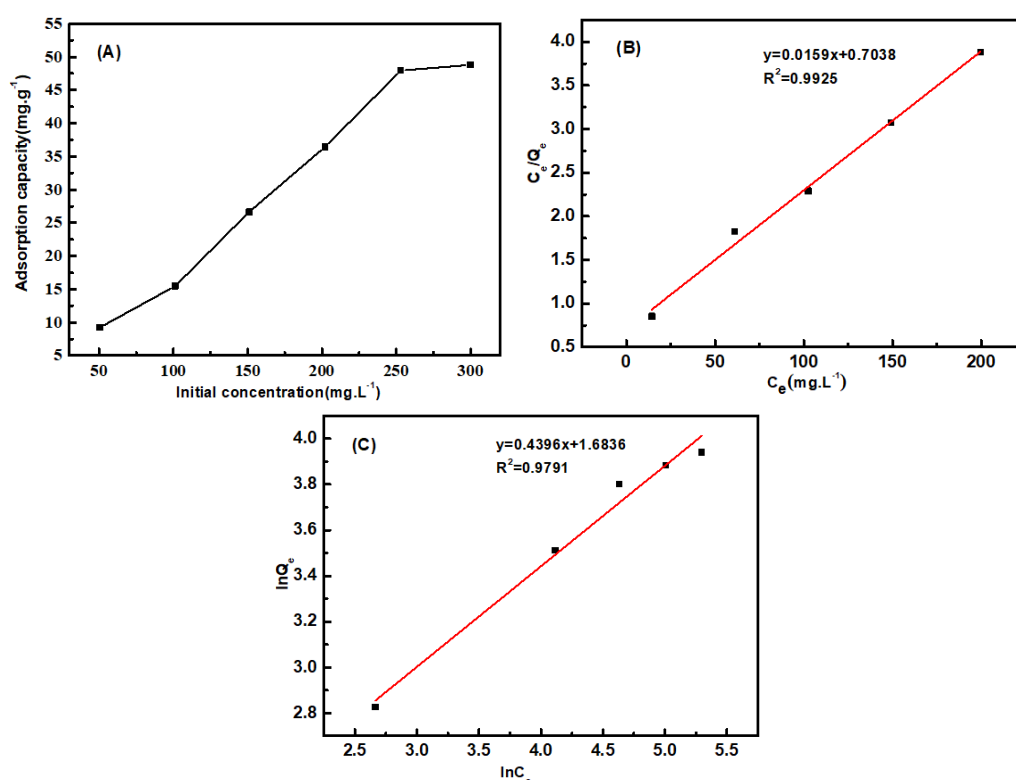


Figure 13. The effect of initial aniline concentration on adsorption capacity (A) and the linearized Langmuir(B) and Freundlich (C) models.

Table 2. Isotherm parameters.

Langmuir			Freundlich		
$Q_m/\text{mg}\cdot\text{g}^{-1}$	K_L	R^2	n	$K_F/\text{mg}\cdot\text{g}^{-1}$	R^2
62.7	0.0226	0.9925	2.2745	5.3851	0.9791

3.6.4. Comparison of Adsorption Capacity with other Adsorbents

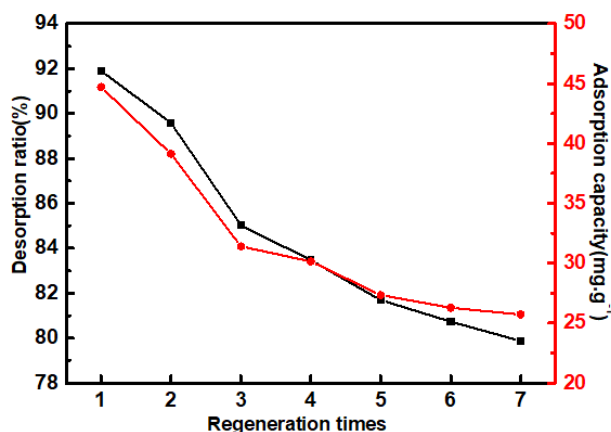
Comparisons between maximum adsorption capacities ($q_{\max}$) of PP-g-(AA-MAH) fibers and other adsorbents for aniline reported in the literature are presented in Table 3. The results show that PP-g-(AA-MAH) fibers exhibits a reasonable capacity for aniline adsorption.

Table 3. Adsorption capacity of various adsorbents for aniline

Adsorbents	$q_{\max}/\text{mg}\cdot\text{g}^{-1}$	Reference
Modified pine sawdust	21.8	[43]
Activated carbon/chitosan composite	22.9	[44]
PAM/SiO ₂	52.0	[45]
Modied ATP	16.1	[46]
Cr-bentonite	21.6	[47]
Modified jute fiber	125	[20]
Edible fungus residue activated carbon	27.1	[21]
PP-g-(AA-MAH) fibers	48.3	This work

3.6.5. Desorption and Regeneration

The regeneration performance of PP-g-(AA-MAH) fibers was investigated using the desorption ratio and adsorption capacity of regenerated fiber, as shown in Figure 14. It can be seen that after seven adsorption-desorption cycles, the desorption ratio of the PP-g-(AA-MAH) fiber still remained above 80% and the adsorption capacity for aniline could still reach 25.72 mg/g, indicating that PP-g-(AA-MAH) fiber had good regeneration-adsorption performance.

**Figure 14.** The desorption ratio and adsorption capacity of regenerated fiber.

4. Conclusions

With PP as the matrix, AA and MAH as functional monomers, PP-g-(AA-MAH) fiber was prepared by suspension grafting and melt-blown spinning technology. The technology can be easily scaled-up to industrial level and the fibers can be used to adsorb aniline. The maximum grafting ratio was 12.47% under the following optimal suspension grafting conditions: grafting time of 3 h, AA: MAH = 0.75, total monomer content of 55%, BPO content of 1.4%, xylene content of 6 mL/g PP, and deionized water content of 8 mL/g PP. Melt flow-rate and thermal analysis indicated that the suspension grafting product had good fluidity and thermal stability, and could be spun by melt-blown process. FTIR and NMR analyses showed that AA and MAH were successfully grafted onto PP fibers. The hydrophilicity of PP-g-(AA-MAH) fiber was enhanced. Therefore, the adsorption for aniline from aqueous solution was fast and the adsorption capacity could reach up to 42.2 mg/g at 45 min and pH = 7. The kinetics of adsorption followed the pseudo-second-order model and the adsorption isotherm data were better fitted by the Langmuir model. Moreover, the fibers had good regeneration performance.

Author Contributions: Conceptualization, Z.L. (Zhouyang Lian) and W.W.; Data curation, J.Z., H.Q. and Z.L. (Zhengwei Luo); Formal analysis, H.Q. and Z.L. (Zhengwei Luo); Investigation, Y.X. and J.Z.; Methodology, Z.L. (Zhouyang Lian), Y.X., H.Q. and W.W.; Writing—original draft, Z.L. (Zhouyang Lian), H.Q., Z.L. (Zhengwei Luo) and W.W.; Writing—review & editing, Z.L. (Zhouyang Lian). All authors have read and agreed to the published version of the manuscript.

Funding: This work was financially supported by the National Natural Science Foundation of China (Grant Number 21676144).

Acknowledgments: W.W. would like to thank the support from the Natural Science Foundation of China (Grant Number 21676144).

Conflicts of Interest: The authors declare no conflict of interest.

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Synthesis of Functional Polypropylene Via Solid-Phase Grafting Soft Vinyl Monomer and Its Mechanism

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Received 14 August 2008; accepted 2 January 2009

DOI 10.1002/app.30179

Published online 17 April 2009 in Wiley InterScience (www.interscience.wiley.com).

ABSTRACT: Solid-phase grafting of a soft vinyl monomer, butyl methacrylate (BMA), onto polypropylene (PP) matrixes with 2,2'-azobisisobutyronitrile (AIBN) as initiator was carried out to enhance the polarity of polymer. Soft vinyl monomer was a novel notion in grafting modification of PP. Effects of swell time, BMA concentration, AIBN concentration, grafting reaction time, and temperature on grafting percentage (G_p) and grafting efficiency (G_e) were examined. The optimal conditions of grafting reaction were obtained: swell time of 60 min, BMA concentration of 6 wt %, AIBN concentration of 0.05 wt %, reaction temperature of 85°C, and reaction time of 2 h. The grafting samples were investigated by such characterization techniques as Fourier transform infrared spectroscopy (FTIR), thermal gravimetric analysis (TGA), and scanning electron microscope (SEM) analysis. FTIR results

indicated that BMA was actually grafted onto PP backbone. TGA results showed that the decomposition temperature increases with addition of BMA into PP backbone. SEM results indicated that the surfaces of PP-g-BMA had a markedly bumpy texture, whereas the pure PP surface was very smooth. Water contact angle results showed that the polarity and hydrophilicity of PP were improved effectively. Compared with the traditional monomer MAH, G_p , and G_e , melt flow rate and mechanical property results all indicated that the soft vinyl monomer had many advantages in the modification of PP. In the end, the mechanism of solid grafting was discussed. © 2009 Wiley Periodicals, Inc. *J Appl Polym Sci* 113: 1803–1810, 2009

Key words: soft vinyl monomer; polypropylene; butyl methacrylate; solid-phase grafting

INTRODUCTION

Polypropylene (PP) is one of the most widely used plastics in automotive, furniture, packaging, and other fields due to its excellent physical and chemical properties, along with low cost. However, PP has its inherent disadvantages such as low surface energy, lack of chemical functionality, and poor compatibility with synthetic polar polymers, which restrict its further applications. This can be resolved through grafting copolymerization, a well-established method for polyolefin modification.^{1–5}

Grafting modification of PP has been widely investigated in past decades to modify the chemical structure of its backbone. Several techniques such as the solution process, melt process, and solid-phase grafting process have been reported, in which solid-phase grafting is most promising.^{6–9} The main advantage of solid-state grafting of PP is that degradation of PP via β -scission reaction at temperature close to 100°C, which would take place extensively when PP is processed in melt state, can be neglected. Most of all, the solid-phase grafting process does not

need excess solvent; the modified PP practically retains its original versatile properties.¹⁰

In most published literature,^{11–15} the general standard of choosing grafting-monomer was grafting percentage of copolymer, but little attention was paid to the physical properties of grafting products. Maleic anhydride (MAH) is a common vinyl monomer widely used to graft onto polypropylene matrix. However, due to its low reactivity and structural confinement, the content of anhydride is very low in grafting copolymer (generally lower than 1 wt %); the crosslinking and degradation often occur during the grafting process, thus leading to a limited degree of functionality and polarity. Besides, the toxicity of MAH also restricts its utility and adds difficulty in the manipulating process.

The soft vinyl monomers used in grafting modification of PP is a novel notion, and an excellent modified polymer may be obtained by introducing them into grafting modification of PP.^{16–19} The term “soft vinyl monomer” means the addition of polymerizable monomers, which, if homopolymerized, would yield a homopolymer having a glass transition temperature (T_g) less than 25°C.¹⁹ Suitable examples of soft vinyl monomer include esters of unsaturated acids and saturated alcohols having from 3 to 15 carbons, such as *n*-butyl acrylate (BA),

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TABLE I
 T_g Values of Some Soft Vinyl Monomers

Soft vinyl monomer	T_g (°C)	Soft vinyl monomer	T_g (°C)
<i>n</i> -Butyl acrylate	−56	2-Hydroxypropyl acrylate	−7
Ethyl acrylate	−24	Methyl acrylate	6
2-Ethylhexyl acrylate	−15	<i>n</i> -Butyl methacrylate	22

2-ethylhexyl acrylate (HEA), and *n*-butyl methacrylate (BMA). The T_g of some soft vinyl monomers in common use are listed in Table I.²⁰ Soft vinyl monomer can polymerize into long flexible branches that regulate the flexibility of PP, reducing the degradation and crosslinking. Moreover, routine consideration of polarity and grafting percentage and grafting efficiency are also satisfied.

Although BMA as a soft vinyl monomer is very reactive, its use for PP grafting has hardly been explored. In this work, grafting copolymer PP-g-BMA was synthesized and characterized. Mechanical property, thermal property, morphology, and hydrophilicity of the products were studied, and very satisfactory results were obtained in comparison with common vinyl monomer MAH system. As far as we know, little attention has been paid to introducing soft vinyl monomer in preparing a grafting copolymer. Therefore it is of significances for academic and industrial researchers.

EXPERIMENTAL

Materials

Porous polypropylene granule (T30s) was supplied by Daqing Petrochemical (Daqing, China). This granular PP was capable of absorbing a larger amount of liquid monomer than could the one in the films, therefore facilitating material feeding. BMA monomer (analytical grade) and 2,2'-azobisisobutyronitrile (AIBN) initiator (analytical grade) were commercially available from the Shenyang Donghua Chemical Company (Shenyang, China). BMA was purged to remove hydroquinone inhibitor, using an inhibitor remover disposable column (Aldrich Chemical) and distilled under reduced pressure prior to use. AIBN was recrystallized twice from methanol. Pure BMA and AIBN were stored at 0°C before use. MAH monomer (chemically pure reagent), commercial methanol, xylene, and acetone were used as received.

Grafting process

The solid-phase grafting process was performed in a three-necked flask equipped with a twin-blade mechanical stirrer operated at 100 rpm, a nitrogen inlet,

and a thermometer. The temperature of the flask was maintained with an accuracy of $\pm 1.0^\circ\text{C}$. First, the reactor was flushed with nitrogen for 3–5 min at room temperature; porous PP granules were then added, followed by a mixed solution of initiator, monomer, and xylene. The solution was then agitated to swelling in an oil bath at the given temperature for 60 min to ensure that this mixture was absorbed by the PP granules. Subsequently the temperature was raised to start the grafting reaction in a nitrogen atmosphere.

Purification of PP-g-BMA

Before characterization, PP-g-BMA samples were purified by Soxhlet extraction in acetone for 12 h to remove BMA homopolymer, xylene, and initiator, then dried *in vacuo* at 50°C for 24 h. After drying, the extracted samples were weighed and re-extracted until the weight of the samples was constant. To prove that BMA homopolymer and other small molecules were extracted, the extracted PP samples were purified in this way: dissolving in refluxing toluene at a concentration of 5% (w/v) and then precipitating in three volumes of acetone at room temperature. The dissolving–precipitating procedure was conducted three times to remove poly (butyl methacrylate), which might be formed in the pores of PP granules. The precipitated samples were dried *in vacuo* at 50°C for 24 h. It was found that the weight was still constant. Therefore it was assumed that monomer and homopolymer were removed completely after extraction process.

Calculation of grafting percentage and grafting efficiency

Grafting percentage (G_p) and grafting efficiency (G_e) were calculated according to the following equations:

$$G_p(\%) = \frac{m_1 - m_0}{m_0} \times 100; G_e(\%) = \frac{m_1 - m_0}{m_2} \times 100$$

where m_1 is sample mass after extraction, m_0 is original mass of PP in sample, and m_2 is total mass of feeding monomers.

Many previous studies^{21–23} have employed the ratio of FTIR characteristic peaks of monomer to internal standard peaks of PP method or titration method to calculate grafting percentage. It must be stressed that the conversion ratio obtained by gravimetry or by two methods mentioned above was in good agreement. Picchioni proved this when he grafted St onto PP.^{24–26} Therefore the gravimetry mode is adopted to calculate grafting percentage in this work.

Gel content of PP-g-BMA

PP might crosslink when a common vinyl monomer is grafted onto it, especially if the grafting percentage is high. If crosslinking occurs, the structures and properties of copolymers are deteriorated to a great extent, which is not what we expected. Thus the crosslinking issue was emphasized in this work. Detecting procedures were addressed as follows: 1.0 g purified PP-g-BMA was added into a given volume of xylene at 80°C and agitated to dissolve for 1 h. As is known, xylene did not dissolve cross-linked PP but pure PP and PP-g-BMA. To further prove that there was no gel in the PP-g-BMA sample, another 1.0 g purified PP-g-BMA was weighed and packed in a copper net with 150 mesh holes. The copper net was weighed beforehand. Then the net was put into a flask with excess hot xylene inside. After refluxing for 1 h, the net was taken out and washed with acetone, then dried in vacuo at 100°C for 24 h. Thereafter, the net was weighed again. The solution was poured into acetone slowly to precipitate PP-g-BMA.

FTIR spectroscopy

FTIR spectra were recorded on a Perkin-Elmer 1760-X FTIR Spectrometer from 4000 to 400 cm^{-1} with a 0.5 cm^{-1} resolution. Samples (0.1–0.2 g) were processed in 100- μm -thick films by compression molding between Teflon-covered aluminium sheets under 1 MPa at 200°C for 5 s.

TGA analysis

TGA was carried out on a Perkin-Elmer RIS Diamond TG-DTA from room temperature to 500°C with a heating rate of 10°C/min under a dry nitrogen atmosphere.

SEM analysis

Samples were fractured under liquid nitrogen and coated with chromium (3 nm) before measurement. The fracture surfaces of samples were observed with a JEOL JSM-6360LA scanning electron microscope operated at 15 kV.

Contact angle measurements

Contact angles of PP and PP-g-BMA were determined using a JC2000C1 contact angle goniometer (Shanghai Zhongchen, Shanghai, China) at ambient temperature. Sample films were processed the same as the ones used in FTIR characterization. All measurements were preformed using the sessile drop method and made with drops of 1 μL distilled water

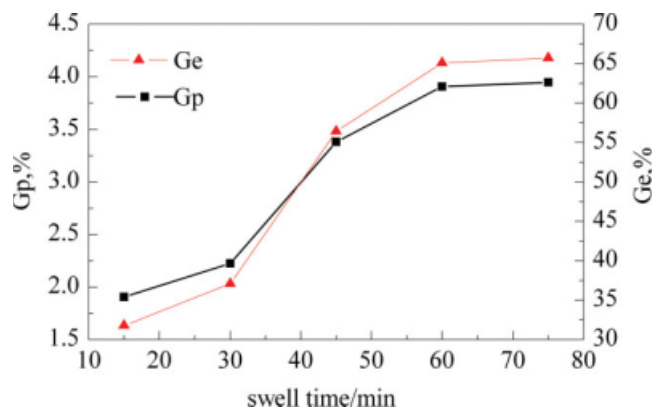


Figure 1 Influence of swell time on G_p and G_e (mass ratio of PP/BMA/AIBN = 100 : 6 : 0.3; reaction time = 2 h; and reaction temperature = 80°C).

after about 15 s. The reported values were an average of eight measurements at various places on same sample.

MFR and mechanical properties measurements

MFR was measured with a Takara X-416 Melt Indexer, (Takara Thermistor Instruments, Japan) Measurement was performed at 230°C with a load of 2.16 kg. Tensile properties of samples were investigated on a universal tensile tester (Instron 1122) using a load of 20.0 kg. Elongation at break and tensile strength were measured at a crosshead speed of 50 mm/min. An average of five test results was calculated and reported.

RESULTS AND DISCUSSION

Experimental conditions such as reaction time and temperature, monomer, and initiator concentration have significant effects on grafting process. Because porous PP granules were first swollen by the solution of BMA/xylene/initiator, the swell time also has important effect on grafting reaction.

Figure 1 shows G_p and G_e versus swell time with diffusion of the solution of BMA/xylene/initiator. The ability of BMA/xylene/initiator solution is to etch and swell the polymer micropore and surface so that it can provide more surfacial areas for absorbing monomer and initiator uniformly. The results indicate that G_p and G_e both reach a constant value after 60 min, although the curves initially increase rapidly. In other words, diffusion equilibrium can be reached within 60 min. Since grafting copolymerization is a diffusion-controlled process, increase of the swell time benefits monomer diffusion into polymer matrix, leading to the presence of sufficient monomer units in grafting zone, which makes a contribution to increase of G_p and G_e . When

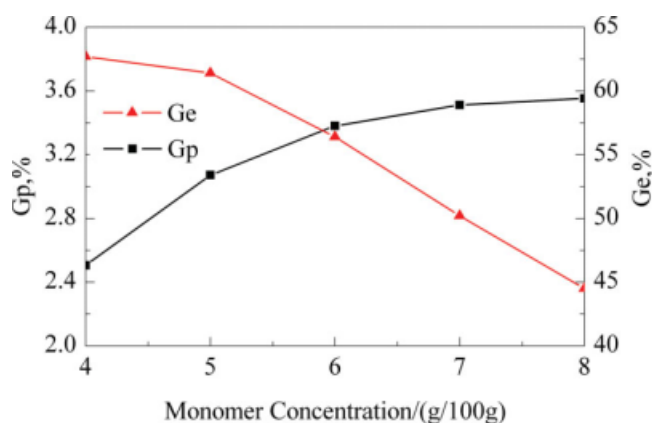


Figure 2 Influence of monomer concentration on G_p and G_e with the swell time of 60 min (other conditions were the same as those in Fig. 1).

the diffusion equilibrium is obtained, G_p and G_e will be constant, independent of time.

The influence of monomer concentration on G_p and G_e is shown in Figure 2. As was said in the Experimental section, BMA and AIBN were first mixed with xylene solution to agitating-swell the polymer matrix. G_p increases quickly as the concentration of BMA increases initially; but when BMA concentration is over 6 wt %, G_p is constant. In contrast, G_e is ceaselessly depressed. A similar trend was observed by Sathe.²⁷ The initial increase in G_p for higher BMA concentration may be attributed to increase of the number of BMA molecules diffusing and reaching the free radical sites on PP backbone. As is well known, PP is a semicrystalline polymer, and grafting copolymerization can only occur at amorphous and inner micropore areas, and their amounts are definite. Furthermore, the free radical sites on PP backbone are initiated by thermal decomposition of AIBN, and the number of active sites is also limited, so G_p will be balanced and G_e will be descended. This result also indicates that the possibility of BMA homopolymer formation increases as the concentration of BMA increases.

Effects of initiator concentration on G_p and G_e are shown in Figure 3. The observed curves are typical for grafting copolymerization reaction via chain transfer. The initial increase in G_p and G_e is caused by an increase in the concentration of initial radicals formed through decomposition of AIBN. Thus, the higher the concentration of initial radicals is, the more the chain transfer to polymer backbone. However, an excess of initiator increases the trend of BMA homopolymerization, which tends to terminate grafting reaction. As a result, G_p and G_e will drop.

Grafting reaction dependence on reaction time is presented in Figure 4. After an initial sharp increase of G_p and G_e , the curves become parallel. This result shows that more than 55 wt % BMA is grafted onto

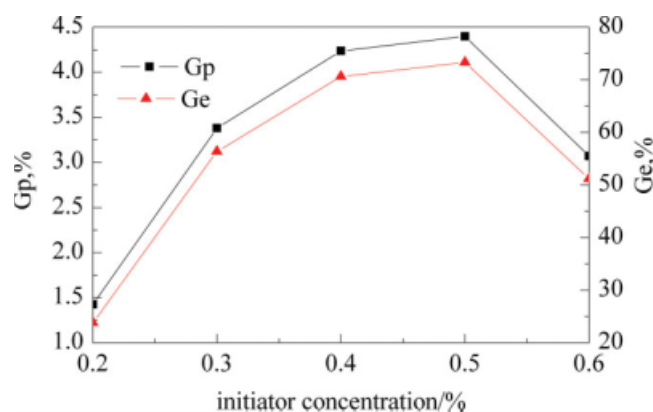


Figure 3 Influence of initiator concentration on G_p and G_e with the swell time of 60 min (other conditions were the same as those in Fig. 1).

PP chains after 2 h, and the rest is consumed in different ways: nonreaction, homopolymerization, or crosslinking.

Effects of temperature on G_p and G_e of grafting BMA onto PP chains were studied with the temperature varying from 75 to 95°C. The results are shown in Figure 5. It can be seen that as temperature increases, G_p and G_e initially increase and pass through a maximum, then begin to decrease. Because of reactive species (PP· radicals) mainly originated from decomposition of AIBN, high reaction temperature will lead to complete decomposition of AIBN and thereby provide enough reactive sites to initiate grafting copolymerization. However, if the temperature is too high, the decomposition rate of the initiator will exceed the grafting polymerization rate and monomers will be initiated to homopolymerize, so G_p and G_e decrease.

Gel content of PP-g-BMA

As stated, the crosslinking issue of PP-g-BMA should be addressed first in this work. The grafted

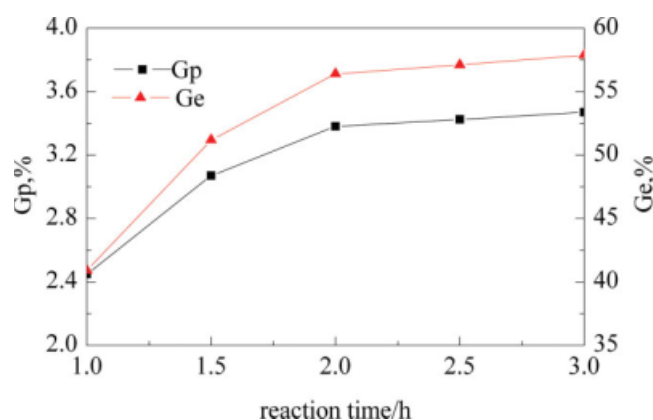


Figure 4 Influence of reaction time on G_p and G_e with the agitating swell time of 60 min (other conditions were the same as those in Fig. 1).

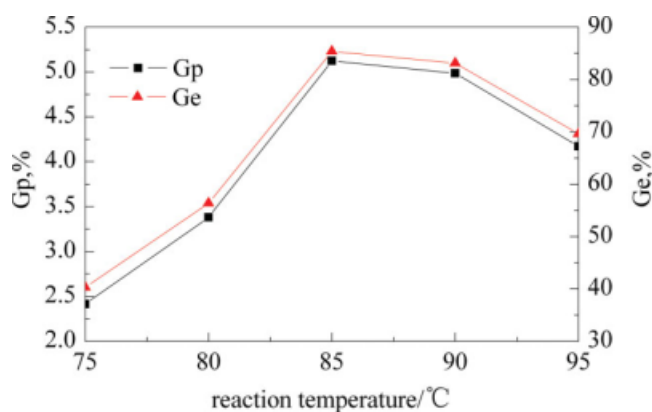


Figure 5 Influence of reaction temperature on G_p and G_e with the swell time of 60 min (other conditions were the same as those in Fig. 1).

polymer was dissolved completely within 30 min without any gel; the weight of the copper net and the purified PP-g-BMA did not change, indicating that BMA as a soft vinyl monomer was grafted onto the PP, and no crosslinking occurred during the grafting process.

FTIR spectroscopy analysis

IR spectra of pure PP and PP-g-BMA are shown in Figure 6. BMA grafted onto PP film was confirmed by FTIR analysis. The presence of a band at 1728 cm^{-1} in the spectrum of the grafted polymer (characterizing the ester group of BMA), which is absent from the spectrum of pure PP, indicates that BMA was grafted onto PP.

Thermal gravimetric analysis

Figure 7 shows TGA results of PP-g-BMA samples with different grafting percentage. It is evident that

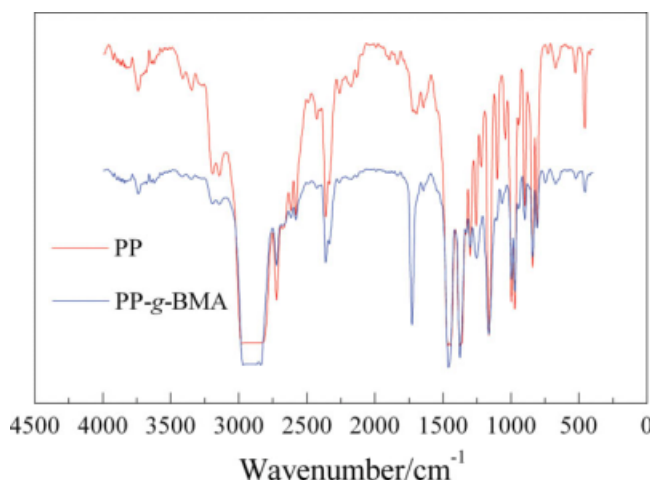


Figure 6 FTIR spectra of pure PP and PP-g-BMA ($G_p = 5.08\%$).

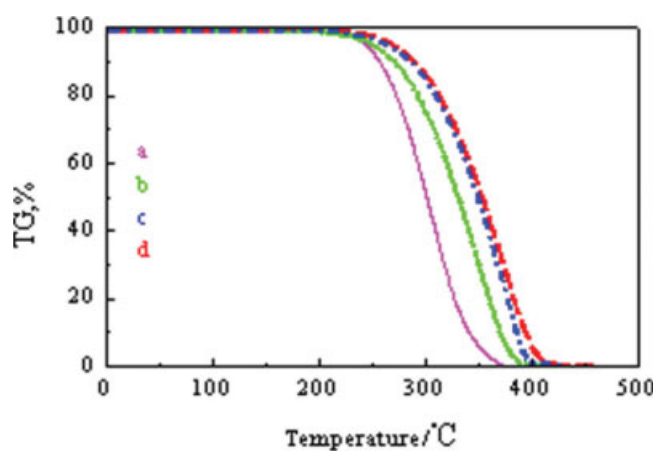


Figure 7 TGA curves of pure PP (a), 1.61% PP-g-BMA (b), 3.43% PP-g-BMA (c), and 5.08% PP-g-BMA (d).

grafted samples have a higher-onset thermal degradation temperature and a lower weight loss at a particular temperature than pure PP. It must be stressed that the more the grafting percentage is, the higher the thermal degradation temperature is. This interesting phenomenon indicates that PP-g-BMA has better thermal stability than pure PP. Clearly, the improved thermal stability of PP-g-BMA results from incorporated BMA branches. This is because the poly(butyl methylacrylate) branch is thermally more stable than tertiary hydrogen atoms on the PP backbone, which results in retardation of degradation rate of PP.

SEM measurements

Figure 8 shows SEM micrographs of cryogenically fractured surfaces of pure PP, as well as different grafting percentages of PP-g-BMA samples. The morphologies of grafted PP [Fig. 8(b,c)] are significantly different from those of pure PP [Fig. 8(a)]. The grafted surfaces show a markedly bumpy texture, whereas pure PP surfaces are very smooth. The bumpy surfaces of grafted layers can be explained by the difference in the swelling ability of xylene/initiator/BMA for amorphous and semicrystalline sites on the PP matrix, resulting in different degrees of grafting in amorphous and micropore regions of PP. Figure 8(b,c) also indicates that grafted molecules are heterogeneously distributed in PP matrix.

Contact angle measurements

Measurement of contact angle between water and a film surface is one of the easiest ways to characterize the polarity and hydrophilicity of a film. When water is applied to the surface, the outermost surface layers interact with water. A hydrophobic surface with low free energy gives a high contact angle with

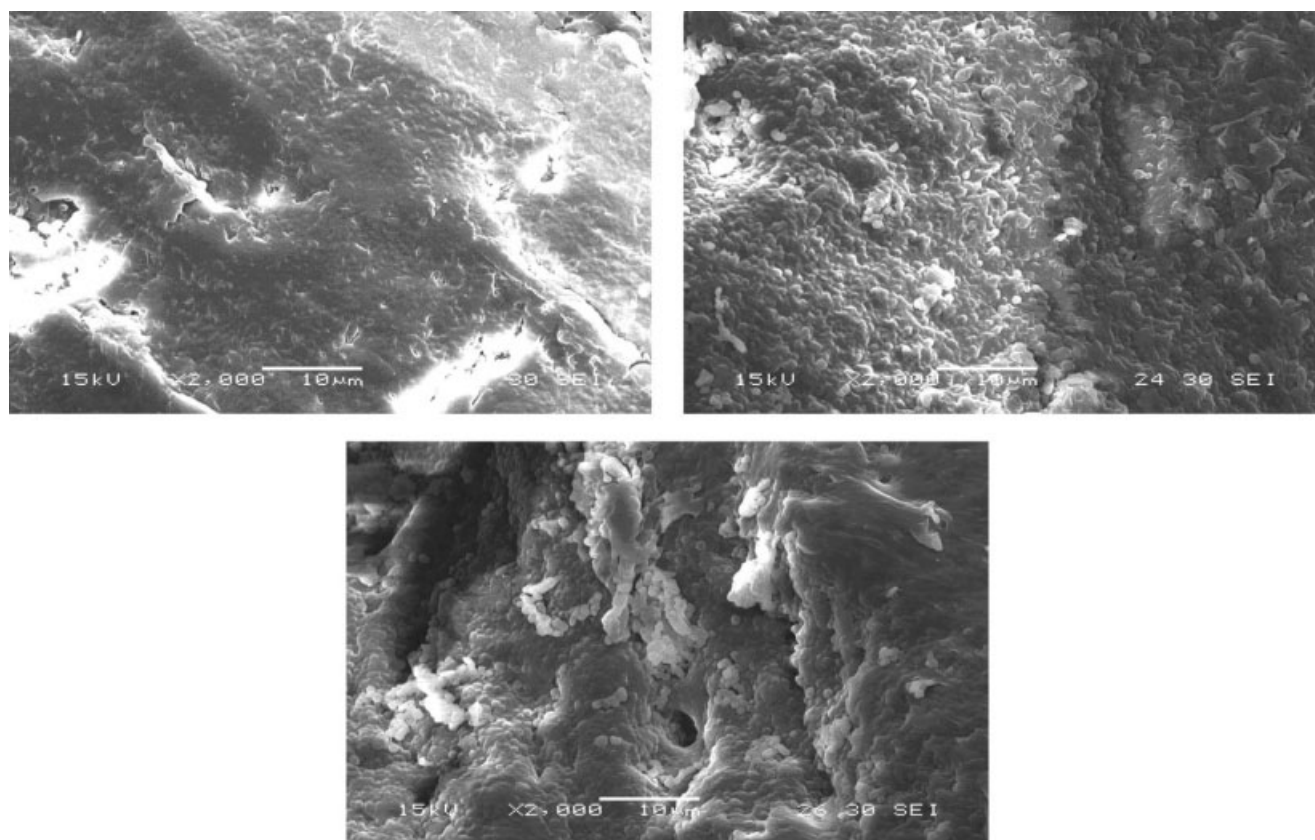


Figure 8 Scanning electron micrographs of pure PP and various grafting percentage of PP-g-BMA: (a) pure PP, (b) $G_p = 1.61\%$, (c) $G_p = 3.43\%$.

water, whereas a wet high-energy surface allows the water droplets to spread, i.e., gives a low contact angle. In this study, the water contact angles of pure PP and PP-g-BMA were measured by contact angle goniometer, and the results are shown in Table II. As is seen, the contact angle decreases from 98° (pure PP) to 75° , indicating that the polarity and hydrophilicity of the polymer were improved efficiently.

Comparison of different monomer systems

As addressed before, MAH is commonly used in grafting modification of PP. Therefore MAH as a common vinyl monomer was chosen in comparison with BMA. The differences of G_p and G_e , MFR, tensile strength, and elongation at break of pure PP, PP-g-BMA, and PP-g-MAH at the same conditions are listed in Table III, respectively. The results in Table III show that G_p and G_e of grafted copolymer

containing soft vinyl monomer, PP-g-BMA, are far higher than those of PP-g-MAH containing common vinyl monomer. Tensile strength and elongation at break of PP-g-BMA are similar to those of pure PP but far higher than those of PP-g-MAH. Interestingly, contrary results are shown in three MFR samples: the MFR of PP-g-MAH is highest, followed by that of PP-g-BMA; the lowest is pure PP. It is easy to understand: BMA as a soft vinyl monomer can improve the polarity of PP and retain its inherent properties, and MAH cannot tune the flexibility of PP backbone and degrades seriously. To further

TABLE III
Comparison of Different Monomer Systems on G_p and G_e , MFR, and Mechanical Properties of Grafting Copolymer

Sample	G_p/G_e (%)	MFR (g·10 min ⁻¹)	Tensile strength (MPa)	Elongation at break (%)
PP	—	3.6	37.02	700
PP-g-BMA	4.25/70.8	6.5	36.79	640
PP-g-MAH ^a	0.45/7.5	22	30.7	20

^a PP-g-BMA and PP-g-MAH were both obtained at the same grafting conditions: PP:AIBN:BMA (MAH) = 100 : 0.3 : 6 (wt %); swell time = 60 min; reaction time = 2 h; reaction temperature = 85°C .

TABLE II
Contact Angles of Pure PP and Different Grafting Percentage of PP-g-BMA

G_p (%)	Pure PP	1.61	4.25
Contact angle ($^\circ$)	98	87	75

explain this phenomenon, we studied the morphology and microstructure of the composites by using different methods that will be reported later.

Mechanism of grafting

In this study, solid-phase grafting is not only a surface grafting process but is also used for grafting onto inner and PP micropores. The mechanisms of grafting resemble that in solution grafting, but not all the same. In solution grafting process, grafting reaction takes place at the molecular level; so does the solid-phase grafting. The decomposition of initiator in the solid phase to form free radicals takes place at the molecular level, but the transport of radicals to PP and initiation of grafting takes place at the solid-liquid interface. The swell process can dif-

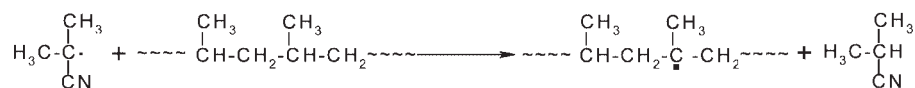
fuse monomer and initiator uniformly into the spherical PP granules, so the whole solid-phase BMA grafting modification of PP can be divided into three steps as follows:

1. By using the porous polypropylene granules, the solution of AIBN/BMA/xylene can be diffused into micropores and surfaces of PP. Thus we can distinguish the transport process of components in the following steps: diffusion in the pores and surfaces of PP granules, absorption in the amorphous phase and micropores, and diffusion in the inner micropores and amorphous phase of microparticles.
2. The grafting mechanism of BMA onto polypropylene backbone with free radical initiation would involve the following steps:

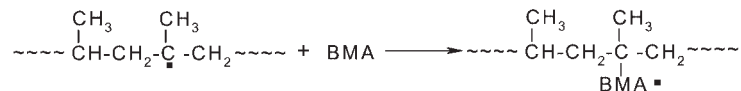
A. Initiator decomposition



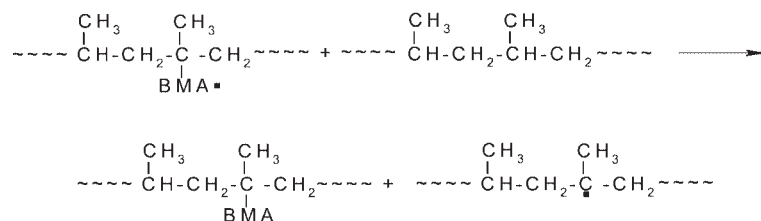
B. Initiation



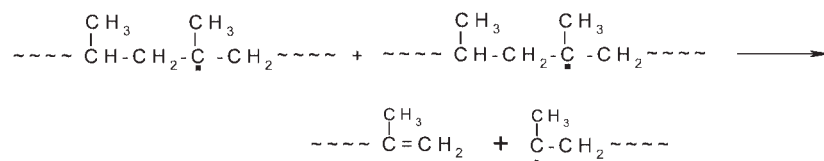
C. Propagation



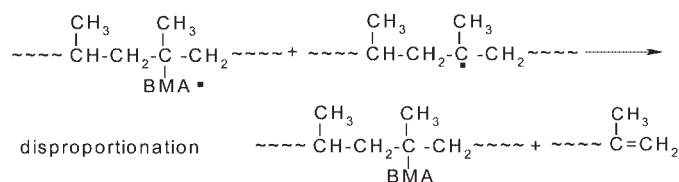
D. Chain transfer

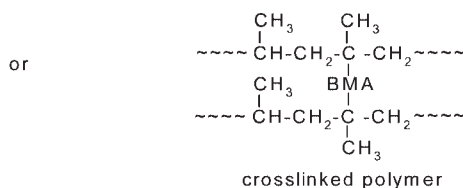


E. Termination



or,





(3) The termination of grafting process and the purification of PP-g-BMA.

CONCLUSIONS

This work concerned the performance of a soft vinyl monomer BMA for free-radical grafting modification of PP in a typical solid phase. Effects of chemical and processing parameters were discussed. The initial decomposition temperature of grafted samples was higher than that of pure PP. The polarity and hydrophilicity of PP were improved effectively. Compared with the common vinyl monomer MAH system, the grafting level, mechanical property, and thermal property of the soft vinyl monomer grafting samples were all satisfactory.

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